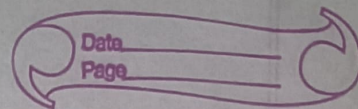


7-march-2026

Intro to Backend Engineering



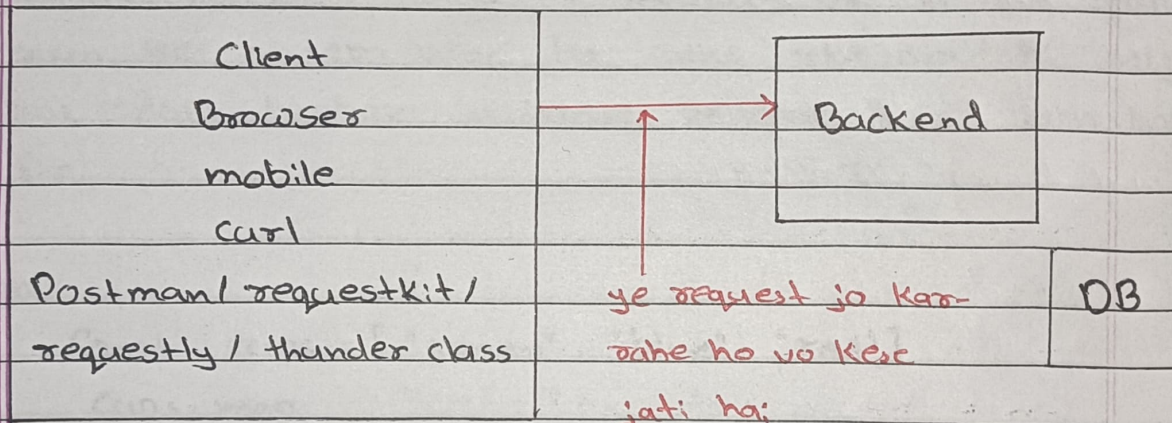
Foundation of backend

Part-I

HTTP methods

(Hypertext Transfer Protocol)

- inko verbs bhi kehte hai
- Jab aap ek browser se backend ke pass request bhejoge to vo request ek kuchh type ki ho sakti hai, vo type hi



aap backend ke under url open karte ho or url blej dete ho or enter karte ho, hamare pass web request aa jati hai, par yesa nahi hota hai

- har jagh alag alag type ki request jati hai jese,

aap ek form bharte ho login id password to alag type ki request jati hai

aap jab kisi ko delete button par click karoke to alag type ki request jati hai

1) GET

- Common request
- to read / retrieve a resource
- there is no side effect of GET
- sirf aap data to read karke lao, data me changes na kardo, db ke under bhi koi changes nahi

IDEMPOTENT

- koi function ko aap call kar rahe ho 1000 time ye ek dam same as calling one time
- agar vo get method hai to uski guaranty honi chahiye ki aap chahe use 1000 bar call karzo chahe ek bar, data same hi milke yala hai
- GET are usually cacheable by browser and CDN (Content Delivery Network)
- GET ke under jese Javascript ya to nextjs framework me, ki aap kisi link pe hover bhi karke ho na, to automatically vo framework aap ke behalf call kar deta hai, jiske karn hme framework fast dikhta hai
- isko prefetching bolte hai
- nextjs bahut aggressively prefetching karke hai
- par sirf GET pe hi kar pata hai

2) POST

- Ek hi kam hota hai iska
create a new resource
- POST is not IDEMPOTENT

- ex, agar 10 request handle hui hai, to 10 bar post hoga, may be user chahta ho 10 bar karna, ha use logic se rok sakte hai, ki 10 bar to nahi karo.

Technically POST ko kabhi bhi IDEMPOTENT nahi hona chahiye

- Calling 10x may create 10 resource.

- POST ke under hamesha data bhejna hi nahi hota, POST khali bhi ja sakti hai. POST me resource read bhi ki ja sakti hai.

- or POST me jab resource create hote hai, tab aap kuchhi response return karte ho jese

⇒ 200 common response hai
all OK, all good

⇒ 201 agar koi resource database me create hua hai
ye status code hai

3) PUT

- It's replace / update a resource completely and sends full Object.

ex, agar database me kuchhi entry ko update karna hai PUT ko use karke, to aap object to pura hi bhejoge jese user ko sirf age overwrite karna hai, par aap bhejoge to sab jese, name, firstname, lastname, email.

- PUT usually IDEMPOTENT kaha jata hai

(PUT ka bhai Patch)

4) PATCH

- It's partially update a resource, send only changed field

Who is more efficient for bandwidth PUT or Patch -
⇒ of course, Patch is the winner

5) DELETE

- remove a resource.
- expect Kiya jata hai ki Delete bhi ~~IDN~~ **IDEMPOTENT** hoga
- ek bar bheja delete karoo bat khatam.

Do Options jo aap use nahi karoge kabhi

1) HEAD

- It's same as get but returns only HEADERS, no body

2) OPTIONS

- Ye Preflight / capability check ke liye use karate hain
- CORS (Cross-Origin Resource sharing)
- ex. jab aap ek request send karate ho na, tab browser request se thoda pehle aap ko ek CORS ki request bhejta hai
- CORS OPTIONS through bheji jati hai
- Browser automatically aapke liye bhejta hai.

Note: HEAD OR OPTIONS Rarely use karate hai

Jab backend se kam ho jata hai tab eiap response bhejte ho.

Is response ke under kya hua vo bheja jata hai.

- jese page na mila to 404 "Page not found" ise hi response ka statusCode khte hai bus ek no. hai.

ex, Responses are grouped in five classes

- 1) Informational responses (100-199)
- 2) Successful responses (200-299)
- 3) Redirection responses (300-399)
- 4) Client error responses (400-499)
- 5) Server error responses (500-599)

HTTP response status codes indicate whether a specific HTTP request has been successfully completed.

Part-II

Command

- node --watch continuous monitor karne ke liye
- node --version current version dekhne ke liye
- npm (node package manager)
- aap kisi or ka code use kar sakte ho npm se

Note:- npmjs.com yaha pe adhere ka code milta hai
and npm owned by Microsoft.

Date _____
Page _____

How to start project professionally

1) npm init

→ Initialize the project

2) fir package ka name node overwrite karne nahi
to node ho jayega direct
package name: (node) mynode

version: (1.0.0)

→ Semver (Semantic Versioning)

it gives the version number

1) MAJOR version when you make incompatible API
changes.

2) MINOR version when you add functionality in a
backward compatible manner.

3) Patch version when you make backward compatible
bug fixes

additional labels for pre-release and build metadata
are available as extensions to the MAJOR.MINOR.
PATCH format.

3) Project Ka description

description: a project to learn typescript

a) entry point main file.

entry point: (index.js) main.js

you can also change in package.json file.
jab chahne tab change kar sakte ho

nahi to direct

npm init -y

Sabhi ko manjuz de dena, bus file generate kardo bus.

ab package.json file hai jo chahi vo install karo.

ex

npm i react (npm install react) both are same

npm i express (npm install express)

TypeScript

- TypeScript is JavaScript with syntax for types and it's strongly typed programming language.

- JavaScript ek loosely types hai ek interpreted language hai

ye ek design hai uske apne ap me ke kuch fayde hai.

ex,

```
function add(a, b) {
```

```
  return a + b;
```

```
}
```

```
console.log(add(1, 2));
```

```
console.log(add('piyush', true));
```

base on + condition

if two number then add

if two string then concatenate.

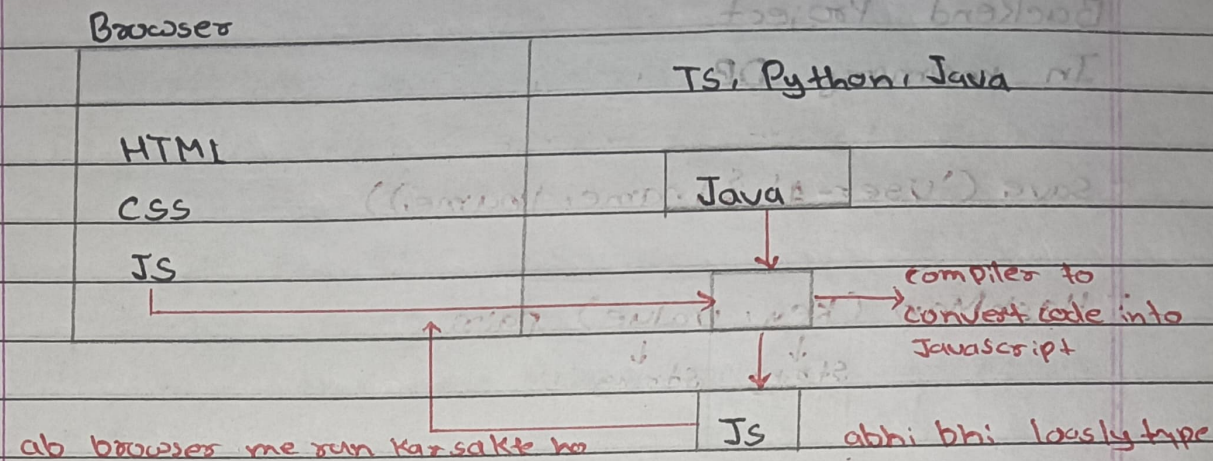
output

3

piyushtrue

Note:- You can't run typescript directly, you have to compile it first.
→ agar typescript me bug hai to use javascript me compile kar hi nahi sakte

Date: _____
Page: _____



You write source code - TS
Compile TS to JS and run
JS file

Koi bhi bug nahi hogi
(Kyuki code pehle java me likha hai)

ex.

```
function add(a: number, b: number): number {  
  a or b dono bhi number data type hai to output bhi hoga.  
  return a + b;  
}
```

→ result bhi number data type hai

```
const result: number = add(3, 4)  
console.log(result)
```

Backend Project

- In memory DB

~~User ka frame user ka first name or last name save karna hai ('user-1', {first name, last name})~~

~~class InMemoryDB {~~

Note:- class is function, but that function acts as a blueprint for creating objects.

Note:- jab bhi ek object banata hai tab hi ek constructor banata hai

- datatype batane ke liye type keyword ka use karna

Backend Project

In memory DB

- Save ('User-1', {fname, lname, email, contact: {mobile, address: {street, pin, country}}})

- HashMap (key, value) pair
↓ ↓
String String

- Ek object, {fname, lname, email, contact: {mobile, address: {street, pin, country}}}

→ datatype batane ke liye type keyword ka use karna
type UserID = String; userID ek string hai

→ shape batana ho data ka tab, interface keyword use karo.

Interface User { user nam data ka shape hai

id: UserID;

fname: String;

lname?: String; → '?' question mark, ye jo

email: String; name na hoto, tabhi bhi

contact: { chal jayega ? matlab optional

mobile: String;

};

address: {

street: number;

pin: number;

country: String;

};

}

Class is a function, but that function acts as a blueprint for creating object

private variable
Can't be access outside class

```
class InMemoryDB {  
    private - db: Map<UserID, User>  
    // ye private variable declare Kiya hai, jiska value map type ka hai, Map<key, value>() => ye jo id nam ka key hai uska type UserID hoga or value ka type User hoga.  
    constructor() {
```

```
        this.-db = new Map<UserID, User>();  
    }
```

→ ID pehle se hi exist hai kinahi
Accessible from anywhere
Output UserID honi chahiye, string

Has it's return boolean and tells if the map contain that specific key

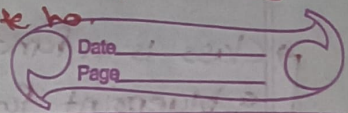
```
    public insertUser (data: User): UserID {  
        if (this.-db.has(data.id)) {  
            throw new Error('User with ID ${data.id} already exists');  
        }  
        this.-db.set(data.id, data);  
        return data.id; // set add new key-value pair to the map. if they already
```

updateData => Kyai kye update karina chahie ho

```
    public updateUser (id: UserID, updateData: Omit<User, 'id'>): boolean {  
        if (!this.-db.has(id))  
            throw new Error('User with ID ${id} does not exists');  
        this.-db.set(id, { ...updateData, id });  
        return true; // Spread operator to merge the data with existing ID.
```

id ko ult: Karke bahar nikal do. matlab usko chhodke baki sab dekho matlab. bhji do.

note! public hoga function object orientent programming me tabhi hi bahar se access kar sakte ho



Retrieve user from the map

```
public getUserById(id: UserID): User {  
    if (!this._db.has(id))  
        throw new Error('User with ID  $\{\{id\}\}$  does  
        string error  $\rightarrow$  not exists');  
    return this._db.get(id)!;
```

}
} $\rightarrow$ to retrieve the user with ki id mili hi na to property key (id) undefined de do

```
const myDb = new InMemoryDB();
```

myDb.I kya ki insertUser public hai

```
myDb.insertUser({
```

id: '1', $\Rightarrow$ string

fname: 'Piyush', $\Rightarrow$ string

email: 'piyush@email.com', $\Rightarrow$ string

contact: { mobile: '99999' }, $\Rightarrow$ string

address: { country: 'In', pin: '147001',

street: '1' }, $\uparrow$ number

});

```
myDb.updateUser('1', {
```

fname: 'piyush',

email: 'piyush@email.com',

contact: { mobile: '99999' },

address: { country: 'In', pin: '147001',

street: '1' },

```
});
```

```
console.log (myDb. getUserById ('1'));
```

output →

```
{
```

```
  'fname' : 'Piyush',
```

```
  'email' : 'piyush@email.com',
```

```
  'contact' : {
```

```
    'mobile' : '99999'
```

```
  },
```

```
  'address' : {
```

```
    'country' : 'In',
```

```
    'Pin' : '147001',
```

```
    'street' : '1'
```

```
  },
```

```
  'id' : '1'
```

```
}
```

```
{
```

```
  'fname' : 'Piyush',
```

```
  'email' : 'piyush@email.com',
```

```
  'contact' : {
```

```
    'mobile' : '99999',
```

```
  },
```

```
  'address' : {
```

```
    'country' : 'In',
```

```
    'pin' : '147001',
```

```
    'street' : '1'
```

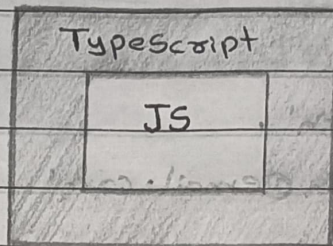
```
  },
```

```
  'id' : '1'
```

```
}
```

TypeScript

TypeScript is a strongly typed programming language that builds on JavaScript, giving you better tooling at any scale.



- JavaScript ke pass, jab aap superpower de dete ho, thodi si, tab vo typescript ban jati hai.

JavaScript se extra kya hota hai fir typescript me

- Types matlab data types: number, string, boolean etc.
- Types jese jese application scale hoti hai uske bugs ko early time pe recognise karne me help karte hai.

Developer Tooling

- it's developer centric tool
- Kabhi kabur kuchh yese bugs hota hai, jo bugs he hi nahi, bus behaviour application ka different ho jata hai

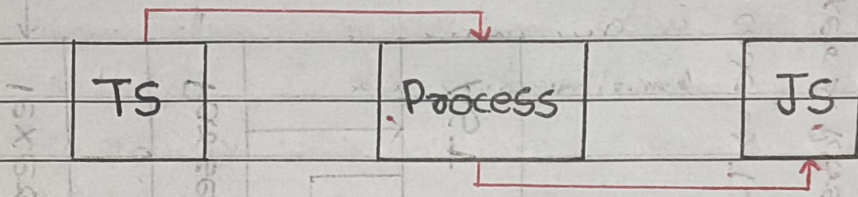
ex.

'Dipali' prints 42

- Printing is not error or bug, behaviour application ka exact nahi hai

Advantages of TypeScript

- TypeScript is an Addon, JavaScript ke uper hi ye bethati hai. Stand alone run nahi kar sakte
- TypeScript actually me run nahi hoti hai.



- TypeScript ko ek process ki jarurat hoti hai run hone ke liye, fir end of day hume JavaScript hi milti hai
- TypeScript JavaScript hi hai bus uspe features laga diye hai
- Jese, type check kar deta hai or uski consistency bhi dekh deta hai code me.

TypeScript code

ex.

```
Function greet (name: string): string {  
    return 'Hello: ${name}';  
}
```

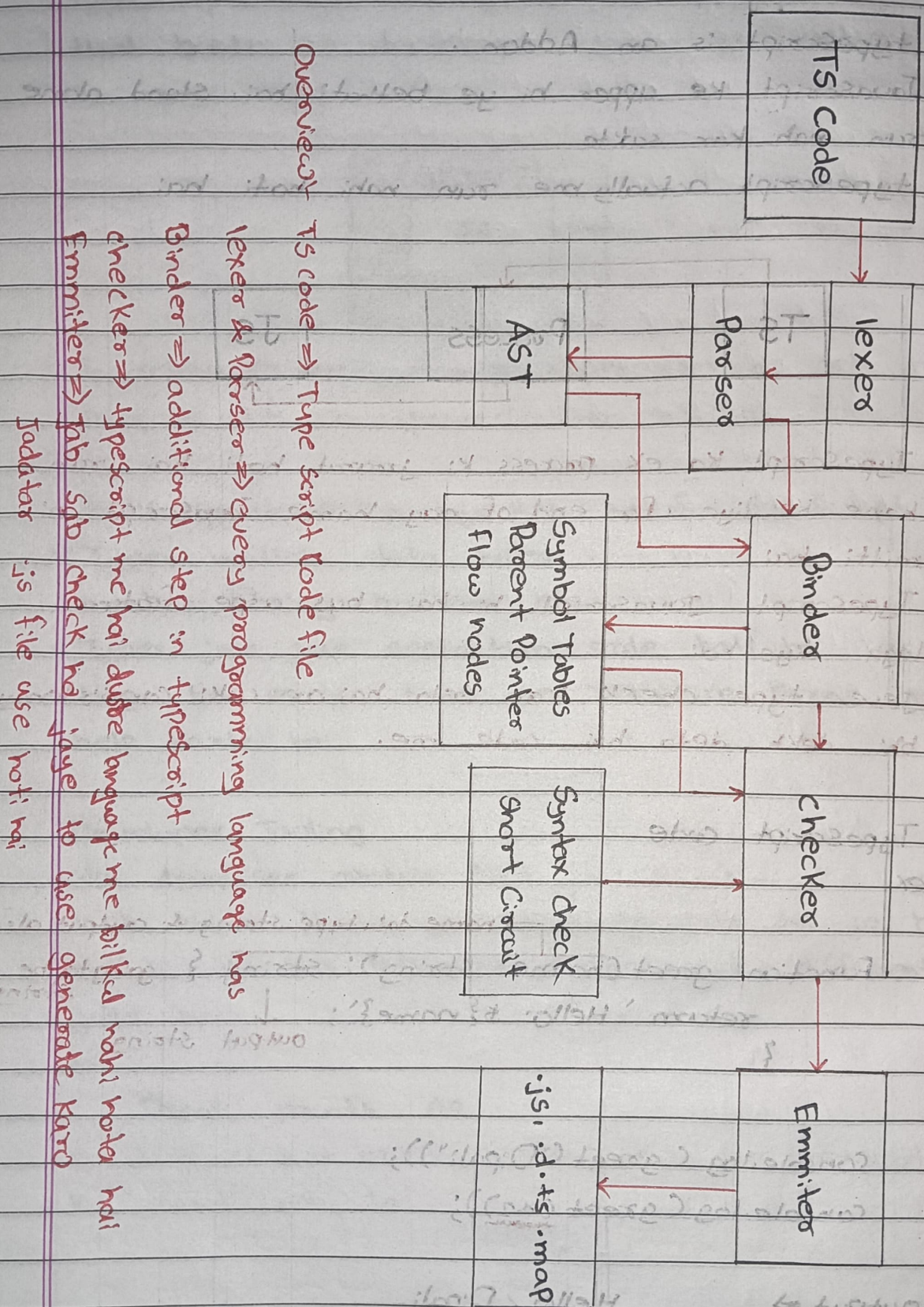
Annotations: 'name' is annotated as 'string' (name data type string). The return type is 'string' (output also going to be string). The return value is 'Hello: \${name}' (output string).

```
console.log (greet ("Dipali"));  
console.log (greet (42));
```

output ⇒

Hello - Dipali
Hello 42

Behind the seen of typescript runtime



In detail

TS Code - Penle Typescript ki file likhi jati hai. jab ko language kam kam ti hai spse pehla step ek lexer ke through leke jao.

lexer - lexer ka kam hai tokenized karna, jese const, let, function, return, jitne bhi keyword wo sare token hote hai. isi ko hi tokenization process bolte hai. Ki aapka pura code ek bar scan hota hai or token ki through jata hai, yahi lexer hai. Ek go through hota hai jese ';', const, function identifier ko address karte hai.

Parser - parser kam hai aap ke liye tree syntax banata hai.

AST (Abstract Syntax Tree)

- AST karta hai ki aapke puri language ke through ek tree banaya jata hai, ki kitne aapke const matlab keywords kitne hai, data ka kya flow hai, ye visualization karne ka kam AST ka hai.

Binder - ye binder AST ko leta hai matlab AST binder ke under jata hai, iske roles

pehle Symbol Tables ko banata hai. Typescript ke under hum kuchh extra symbol add karte hai jese name: string fir undefined jo definition bahnti hai. unhi symbol ko symbol table me banaya jata hai.

fir Parent Pointer jese Ast ek tree hai, tree me dom bhi aana padta or upper bhi jana padta hai, jab niche aa jate ho tab upper wale parent ko node ka hisab rakhta hai wapis jane ke liye.

Flow nodes matlab if-else ke flow ho gye.

jese ye nahi huva to ye karlo.

or Symbol tables, Parent Pointer or Flow nodes

ye sara kam binder ka hota hai.

ye sara ho jane ke bad checker pe jaya jata hai.

checker - ye checker do bar pure code ko go-through

kar ta hai. isliye kabhi kabar compile hone me

time lag jata hai.

iske under basically har ek function ko, har ek

node pe jate hai or check karte hai, ki uska

data type jo available tha vo kahi kisi or ko

alot karte smay us symbol ko some symbol

ko kuchh or data type ya value allot karte smay

data type to change nahi ho gaya.

yaha pe kuchh concept aate hai, jese.

Short Circuit matlab pehle sahi kuchh define ho

gaya value, ko to jada check na karo.

Syntax Check yaha pe syntax ki strict checking

hoti hai.

or ye sara kam checker ke under hota hai.

do pura kam ho jane ke pass emitter ke pass

jata hai.

Emitter - Emitter ka kam itna hi hai ki, file ko generate

karta hai or striping karta hai.

Jo type script ne types lagaye hai unhe finally

remove karna.

or ye emitter ka kam node karta hai

node typescript sirf process nahi karta, vo

uske types ko stripout karta hai.

or ye emitter se hota hai.

nodejs

- nodejs javascript ka runtime environment hai
- pehle jo javascript ko execute karta tha pura engine vo browser ke under tha.
- to jab engine browser ke bahar nikala.
Not just JS on the server.
or vo Google ka V8 engine tha.
- par javascript to sirf code run karti hai, uske pass console hai, na networking capability hai. peduon request jese, ye kisi ko ping nahi hai or times ko bhi DOM bhi nahi hai.
- to engine se sat kuchh attachment lage un sabhi ko fir nodejs karte hai.

nodejs

		Google V8	
		Engine	
		Libuv	asnc I/O
		layer	single thread

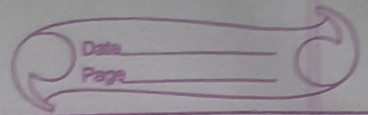
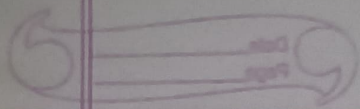
- sab se pehle google ke v8 engine ke upper ek layer lagai gai, one of the most popular library, libuv.
- libuv library aap ko asnc I/O layer, ye vo layer hai jo network se baat kar sakti hai Input, output kar sakti hai, OS se bhi connect karti hai.
- or ye puri chiz bunte nodejs banti hai.

Core Component

1) V8 Engine (google design it)

- Is ka kam hai ki aapke JS code ko ~~me~~ machine code me convert karta hai, use JIT compilation bolte.
JS $\Rightarrow$ Machine code (JIT compilation)
- **JIT $\Rightarrow$ Just In Time.**
(utna compile karlo jitne ki abhi jazaout hai)
- par typescript pura compile karti hai, pura hi Javascript ~~through throw~~ karti hai.
- Javascript ko interpreted languages ko compile nahi kar sakte, or compile language fast hoti hai.
- Interpreted language slow hoti hai, kyu ki code usi time ~~reack~~ check hoke execute, error hoke milti hai esliye.
- ~~Compile me ye software hai sidha run kardo~~
- Compile me sub check ho gaya hai, aap ke pass sidha software hai run kardo, precheck hota hai isliye fast hai.
- OR iska kam hai, aapke heap or call stack ko manage karna, memory bhi management karta hai or queue ko bhi yahi karte hai.
- par inhe abhi bhi kuchh nahi pata files, network or timers ke bare me.

note:- node pura hai VR ek bus part hai.



2) LIBUV (Pure C Library)

~~- ye event~~

- ye event loop, thread pool, OS level async I/O, DNS lookup sub deti hai.

3) OR nodejs Ki bindings (C++)

- jese ek `fs.readFile()` hai, JavaScript ne is File system ko call kar diya, par ye read file LIBUV karega, OR indono ke bich je kam karaye ga unhe bridge hai.

- aap code likho je java script me, but bridge bat karega LIBUV se C ya to C++ me, OR fir vo data bridge me lake dunga.

- JavaScript single thread language hai, par baaki ke thread LIBUV provide karta hai.

OR ye 3no pura ka pura jab bind bind karte ho use node js kehte.